

# BMKG STRONG MOTION ANALYZER

Laporan Analisis Strong Motion

## 1. RINGKASAN GUNCANGAN TERKUAT

Stasiun PGJM \_ 2026-02-05 18\_04\_34 UTC mencatat nilai percepatan tanah maksimum (PGA) sebesar 0.740023 m/s<sup>2</sup> atau 74.002 Gal pada komponen HNN. Berdasarkan klasifikasi SIG-BMKG yang digunakan oleh BSMA, hasil tersebut termasuk Skala II (DIRASAKAN) dengan rentang intensitas III-V.

|               |                             |
|---------------|-----------------------------|
| Network       | IA                          |
| Station       | PGJM                        |
| Location      |                             |
| Sampling Rate | 100.000 Hz                  |
| Start Time    | 2026-02-05T18:04:34.193130Z |
| End Time      | 2026-02-05T18:12:34.183130Z |
| Channels      | HNE, HNN, HNZ               |

### SKALA II SIG-BMKG | DIRASAKAN

Warna: Hijau | Rentang intensitas: III-V

Dampak: Dirasakan oleh orang banyak tetapi tidak menimbulkan kerusakan. Benda-benda ringan yang digantung bergoyang dan jendela kaca bergetar.

### INSTRUMENTAL INTENSITY (MMI): V | PERCEIVED SHAKING: MODERATE

Potensi Kerusakan: Very light | Peak Acc Baseline: 6.2 %g | Peak Vel Baseline: 4.7 cm/s

Nilai Rekaman Terukur: PGA = 7.546 %g | PGV = 2.292 cm/s (Skala USGS ShakeMap / Worden et al., 2011)

## 2. PARAMETER STRONG MOTION

| Channel | PGA (Gal) | PGV (cm/s) | PGD (cm) | Arias (m/s) | D5-95 (s) | SIG        |
|---------|-----------|------------|----------|-------------|-----------|------------|
| HNE     | 52.089    | 1.6654     | 0.1148   | 0.028777    | 22.77     | II (III-V) |
| HNN     | 74.002    | 2.2920     | 0.1939   | 0.049309    | 20.62     | II (III-V) |
| HNZ     | 17.507    | 0.8560     | 0.0983   | 0.004049    | 24.33     | II (III-V) |

### Parameter komponen terkuat:

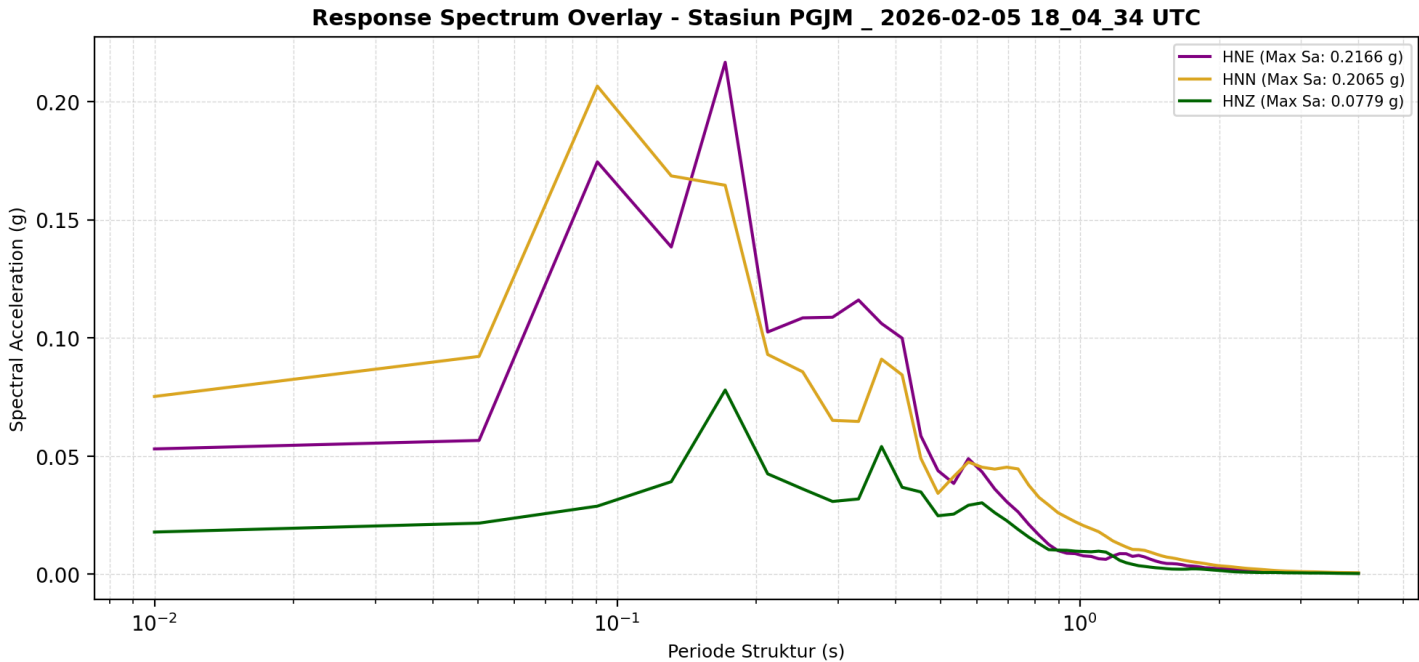
PGA = 0.740023 m/s<sup>2</sup> (74.002 Gal); PGV = 2.2920 cm/s; PGD = 0.1939 cm; Arias Intensity = 0.049309 m/s; D5-95 = 20.62 s.

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## 3. RESPONSE SPECTRUM

Kurva response spectrum menunjukkan percepatan spektral sebagai fungsi periode struktur berdasarkan hasil pemrosesan response spectrum pada pipeline BSMA.

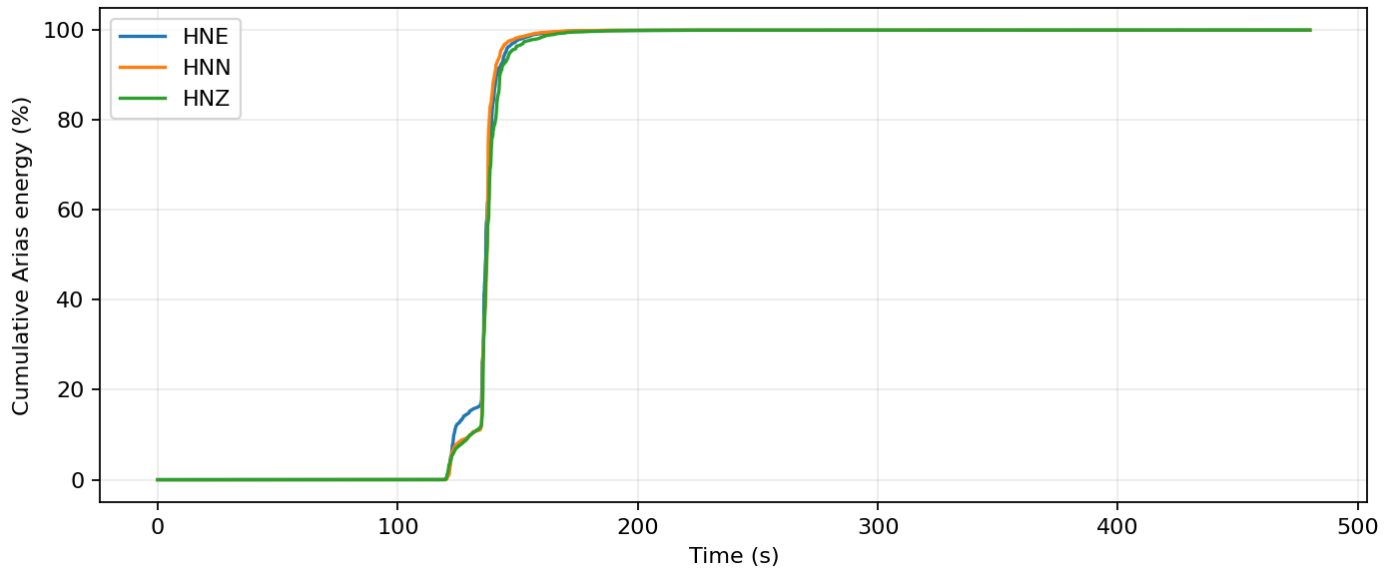


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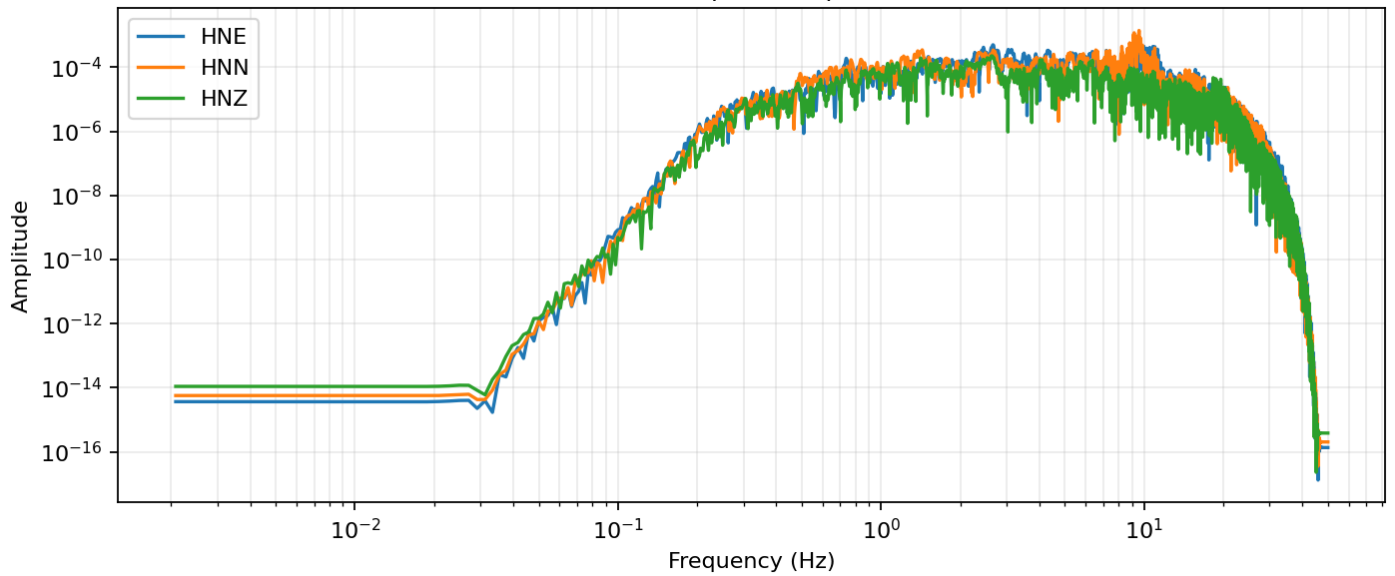
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## 4. HUSID PLOT DAN FOURIER AMPLITUDE SPECTRUM

Husid energy curve



Fourier amplitude spectrum



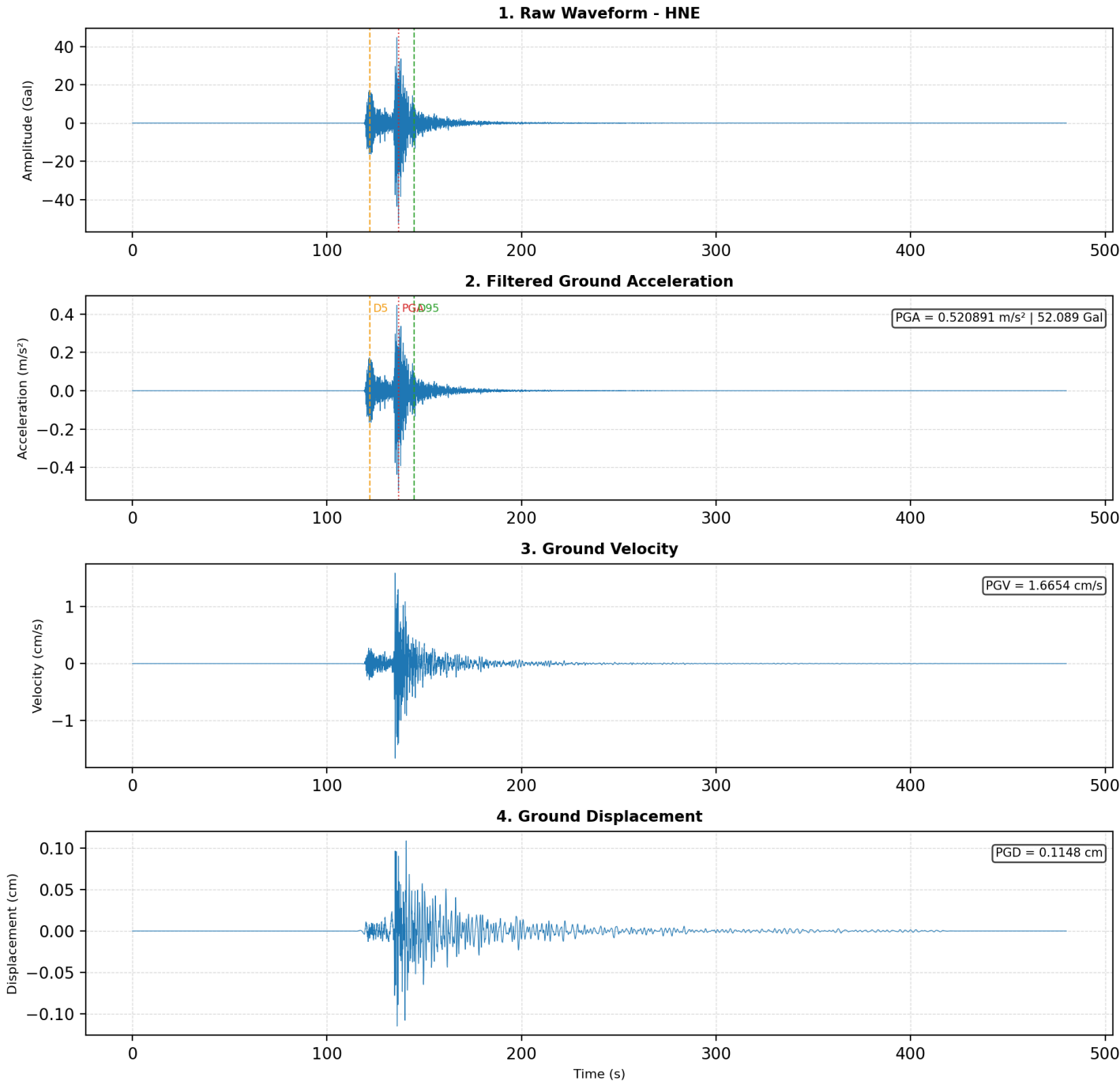
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## 5. WAVEFORM ANALYSIS - HNE

PGA: 52.089 Gal | PGV: 1.6654 cm/s | PGD: 0.1148 cm | D5-95: 22.77 s

Analisis Kinematik Ground Motion  
Stasiun: PGJM \_ 2026-02-05 18\_04\_34 UTC | Channel: HNE



# BMKG STRONG MOTION ANALYZER

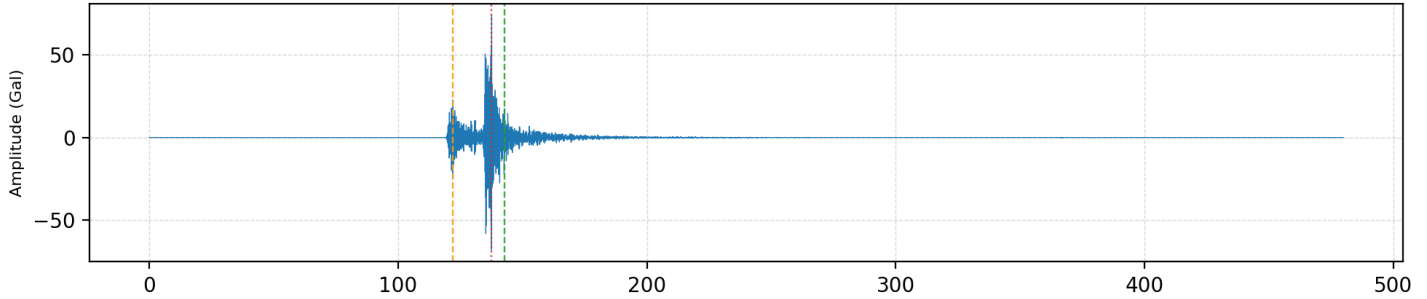
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## 5. WAVEFORM ANALYSIS - HNN

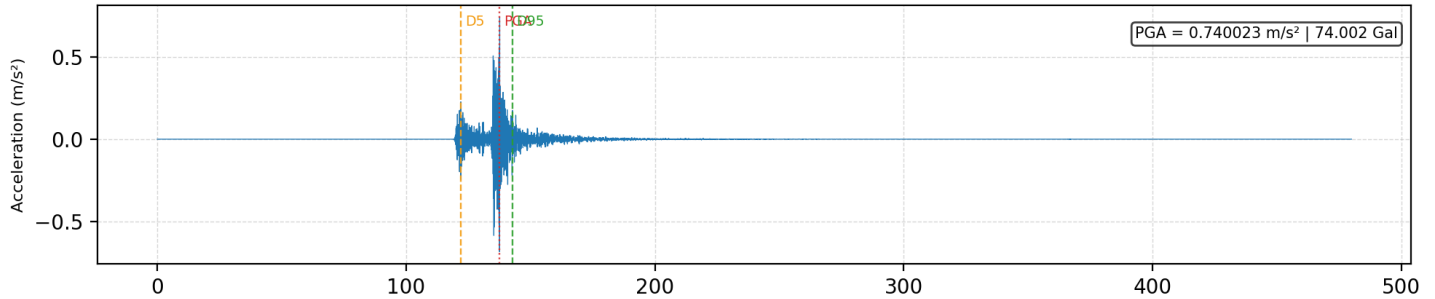
PGA: 74.002 Gal | PGV: 2.2920 cm/s | PGD: 0.1939 cm | D5-95: 20.62 s

**Analisis Kinematik Ground Motion**  
**Stasiun: PGJM \_ 2026-02-05 18\_04\_34 UTC | Channel: HNN**

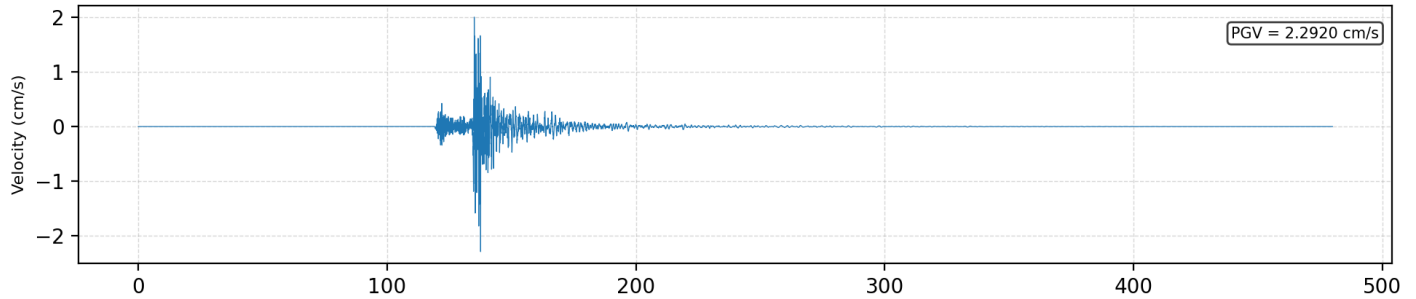
1. Raw Waveform - HNN



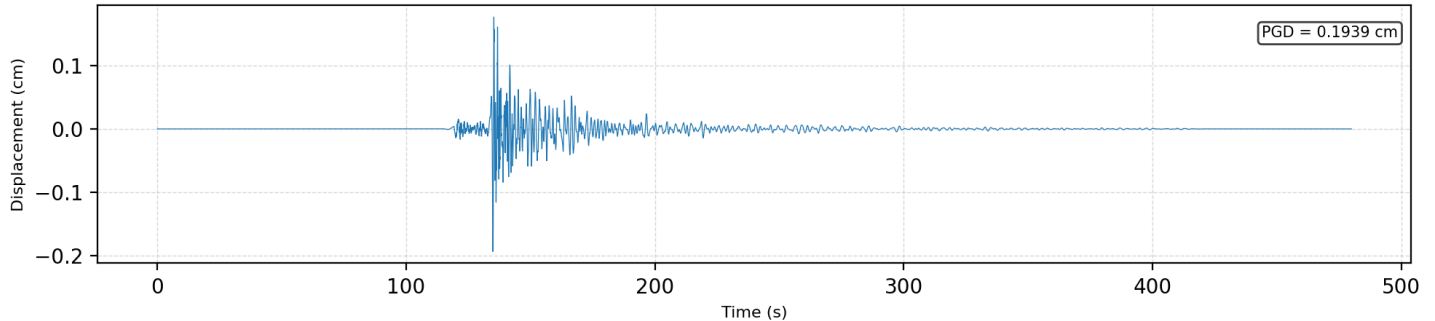
2. Filtered Ground Acceleration



3. Ground Velocity



4. Ground Displacement



# BMKG STRONG MOTION ANALYZER

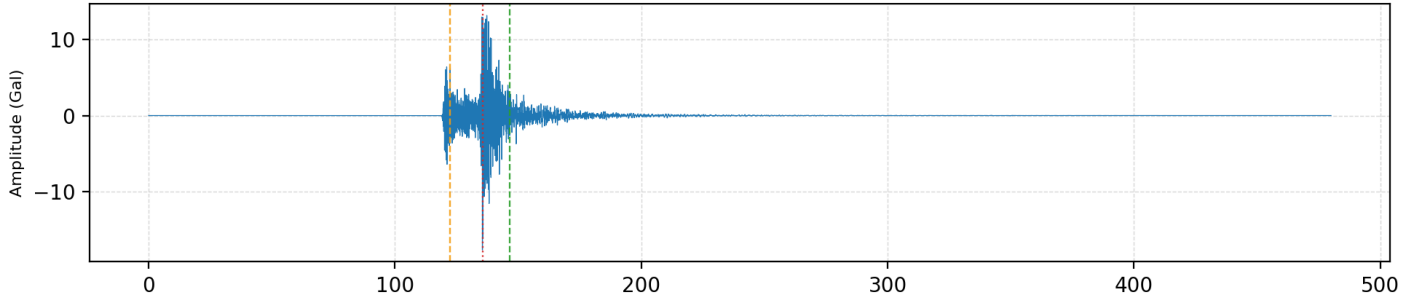
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## 5. WAVEFORM ANALYSIS - HNZ

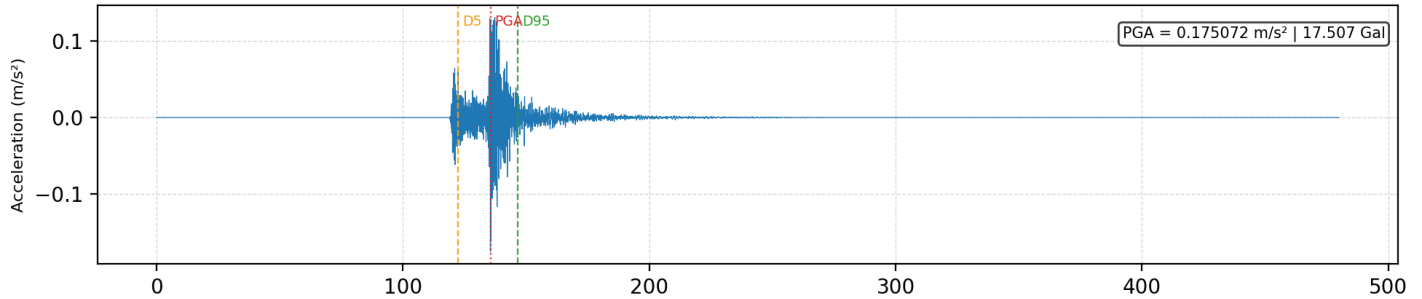
PGA: 17.507 Gal | PGV: 0.8560 cm/s | PGD: 0.0983 cm | D5-95: 24.33 s

**Analisis Kinematik Ground Motion**  
**Stasiun: PGJM \_ 2026-02-05 18\_04\_34 UTC | Channel: HNZ**

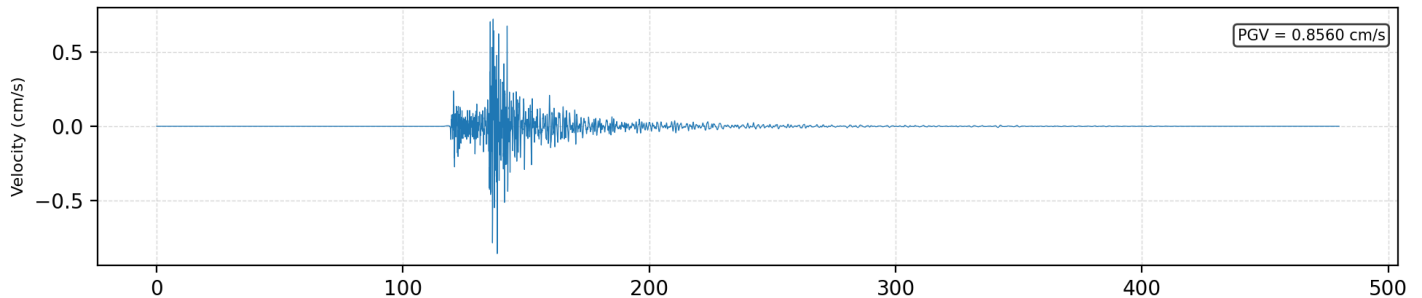
1. Raw Waveform - HNZ



2. Filtered Ground Acceleration



3. Ground Velocity



4. Ground Displacement

